



### Current Academic appointments

- 2025– **Associate Professor of Computer Science**, *Department of Economic Sciences*, University of Bergamo, Bergamo, Italy
- 2026– **Rector's Delegate for Artificial Intelligence**, University of Bergamo, Bergamo, Italy

### Past Academic appointments

- 2022–2025 **Researcher in Computer Science**, *Department of Economic Sciences*, University of Bergamo, Bergamo, Italy
- 2020–22 **Associate Professor of Operational Research**, *Department of Mathematical Sciences*, University of Southampton, Southampton, UK
- 2016–20 **Lecturer (Level 5, equiv. to Lecturer B) in Operational Research**, *Department of Mathematical Sciences*, University of Southampton, Southampton, UK
- 2019 **Visiting Professor**, *Université Paris Dauphine*
- 2018 **Visiting Professor**, *Politecnico di Milano*
- 2018 **Visiting Professor**, *Université Paris Dauphine*
- 2012–16 **Postdoctoral researcher**, *Lehrstuhl II für Mathematik (Department of Mathematics II)*, RWTH Aachen University, Aachen, Germany, Discrete Optimization group headed by Prof. Arie Koster
- 2011–12 **Postdoctoral researcher**, *Dipartimento di Elettronica e Informazione (Department of Electronics and Computer Science)*, *Politecnico di Milano, Italy*, Operations Research group headed by Prof. F. Maffioli
- 2010 **Visiting scholar**, *Tepper School of Business, Carnegie-Mellon University, Pittsburgh, US, Pennsylvania*, Supervisor: Prof. F. Margot

### Qualifications

- 2022 **National Scientific Qualification (Abilitazione Scientifica Nazionale) as Full Professor - sector "01/A6 Ricerca Operativa" (Operations Research)**
- 2021–23 **Fellow of the Alan Turing Institute, UK**
- 2021 **National Scientific Qualification (Abilitazione Scientifica Nazionale) as Associate Professor - sector "01/B1 Informatica" (Computer Science)**
- 2020 **National Scientific Qualification (Abilitazione Scientifica Nazionale) as Associate Professor - sector "01/A6 Ricerca Operativa" (Operations Research)**
- 2017– **Fellow of the Academy HE (formerly known as Higher Education Academy) (awarded: 15.12.2017)**, awarded in recognition of attainment against the "UK Professional Standards Framework" for teaching and learning support in higher education  
Recognition reference: PR139371
- 2008–11 **PhD in Information Technology (awarded: 16.02.2011)**, *Dipartimento di Elettronica e Informazione (Department of Electronics and Computer Science, Operations Research Group)*, *Politecnico di Milano*, Advisor: Prof. E. Amaldi  
Thesis: *On coordinated cutting plane generation and mixed integer programs with nonconvex 2-norm constraints.*
- 2005–07 **MSc in Engineering of Computing Systems (awarded: 20.12.2007)**, *Politecnico di Milano*, Grade: 110/110 cum laude  
Thesis: *Hyperplane clustering and piecewise linear model fitting.*
- 2002–05 **BSc in Engineering of Computing Systems (awarded: 28.09.2005)**, *Politecnico di Milano*, Grade: 110/110 cum laude  
Thesis: *New constructive heuristics for finding minimal fundamental cycle bases in graphs.*

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## Scientific and professional memberships

- 2026– **Member of the ELLIS Society e.V.**, European Laboratory for Learning and Intelligent Systems (ELLIS), ELLIS Unit Milan
- 2026– **Member of the Unione Matematica Italiana (UMI)**

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## Research grants

- 2023–25 **Co-Investigator of the project "New skills for the Public Administration: human-centered AI in the training of public employees" – "Nuove competenze per la PA: human-centered AI nella formazione dei dipendenti "pubblici (FAI - PA), funder: Dipartimento della funzione pubblica, budget: 99.555,50 EUR**
- 2023–25 **Principal Investigator of the project Space-time graph convolutional neural networks (ST-GCN): learning and predicting time-varying graph relationships with applications to the air transportation sector, funder: University of Bergamo, budget: 64,800 EUR**
- 2023–25 **Investigator (and Local-Unit Leader) of PRIN-PNRR Project HEXAGON: Highly-specialized EXact Algorithms for Grid Operations at the National level, funder: EU's Next-Generation EU initiative and Italian Ministry of University and Research (MUR), budget: 232,300 EUR**
- 2022–22 **Investigator of Project 7316614: Ad Hoc Training On Stochastic Programming & Optimisation, funder: Sainsbury's, UK, budget: 6,400 GBP**
- 2020–22 **Recipient of a Royal Society and China National Science Foundation International Exchange Grant, funder: the UK's Royal Society and China's National Science Foundation, budget: 12,000 GBP**
- 2020–21 **Investigator of the "Economic Ageing of Transformers" project, funder: the National Grid, budget: 446,814 GBP**
- 2018–19 **Recipient of the Paris Dauphine Visiting Professor Grant, funder: Université Paris Dauphine, budget: 2,900 GBP**
- 2018–19 **Recipient of the ESRC Impact Acceleration 'Next Generation Services' fund, funder: University of Southampton, budget: 3494.40 GBP**
- 2017–18 **Recipient of the Paris Dauphine Visiting Professor Grant, funder: Université Paris Dauphine, budget: 2,900 GBP**
- 2017–18 **Recipient of the SIRDF-FBLA Strategic Interdisciplinary Research Development Fund 2017/2018 for the project: W-GREENBELT: Workshop on GREEN and BluE Logistics and Transport, funder: Faculty of Business, Law and Arts, University of Southampton, budget: 6,789 GBP**
- 2016–17 **Recipient of the SIRDF-FHMS Strategic Interdisciplinary Research Development Fund 2016/2017 for the project Hierarchical optimization techniques for forecasting, funder: Faculty of Social, Human and Mathematical Sciences, University of Southampton, budget: 6,959 GBP**
- 2015–16 **Recipient of the RWTH Start-Up grant for early-career researchers for the project From the simplex method to cutting plane algorithms: bound-optimal cuts and bound-improving pivoting rules, funder: RWTH Aachen University, budget: 20,000 EUR**

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## Participation to research projects and consultancy activities

- 2022–22 **Ad Hoc Training On Stochastic Programming & Optimisation, funded by Sainsbury's, UK**
- 2020–21 **Economic Ageing of Transformers, funded by the National Grid, UK**  
Development of optimization algorithms for assessing the economical impact of updating the ratings of transformers
- 2018 **UK Met Office, Exeter, UK**  
Development of optimization algorithms for personnel rostering subject to fairness and cyclicity constraints.
- 2014–16 **Robust optimization of power networks, funded by BMWi, Bundesministerium für Wirtschaft und Energie (Federal ministry for economy and energy), Germany, industrial partner: ProCom GmbH, university: RWTH Aachen University**
- 2013–16 **VINO: Virtual Network Optimization, funded by BMBF, Bundesministerium für Bildung und Forschung (Federal ministry for education and research), Germany, university: RWTH Aachen University**
- 2012–13 **Robust Mixed-Integer Linear Programming, funded by Google Inc., university: RWTH Aachen University**
- 2011–12 **Planning and operation of IP networks with low energy consumption, funded by Lombardy Region (Italy), within the project "Green ICT & ICT for Green", university: Politecnico di Milano**

- 2011–12 **CESI (Centro Elettronica Sperimentale Italiano, Italian Center for Experimental Electronics)**, Milan, Italy  
Development of Mixed-Integer Linear Programming techniques to solve optimal power flow, outage minimization, and unit commitment problems in the Italian power network.
- 2011 **Coordinated cutting plane methods for integer linear programming**, funded by MIUR, Ministero dell'Istruzione, dell'Università e Ricerca (Ministry for education, university, and research), Italy, university: Politecnico di Milano
- 2008–10 **DECEMbRIA: decisions in health-care emergencies**, funded by Lombardy Region, Health-care Directorate-General (Italy), partner: 118 Emergency Service operations center, university: Politecnico di Milano

## Industrial collaborations within student supervisions

- 2016– **Collaborations with the following companies**, Airbus, British Telecom, Babcock Aviation, Babcock Land, Gousto, De La Rue, Royal National Lifeboat Institute, Hampshire City Council (UK), Johnsons Controls, Microlise, Movement Strategies

## PhD dissertation

- [D1] **On coordinated cutting plane generation and mixed integer programs with nonconvex 2-norm constraints**, Dipartimento di Elettronica e Informazione, Politecnico di Milano, 2011

## Edited volumes

- [E1] **Computational Logistics: 8th International Conference, ICCL 2017, Southampton, United Kingdom, October 18-20, 2017, Proceedings**, with T. Bektas, T. Martinez-Sykora, S. Voß, Lecture Notes in Computer Science, Springer [Publisher](#), volume 10572

## Chapters in books

- [C2] **Definitional questions in a historical perspective**, “Artificial Intelligence: a research agenda (Intelligenza artificiale: un’agenda di ricerca)”, M.F. Murru, ed., Università degli studi di Bergamo, 2024
- [C1] **A Unified Framework for Multistage Mixed Integer Linear Optimization**, with S. Bolusani, T. Ralphs, S. Tahernejad, “Bilevel optimization: advances and next challenges”, Dempe, S., Zemkoho, A., eds., Springer, 513–560, 2020 [Publisher](#) · [arXiv](#)

## Journal papers

- [J29] **Bridging theory and practice: Efficiently solving the unit commitment problem in production cost models**, with P. Bunnak, S. Galelli, Energy, 322 (2025), 135454 [Publisher](#)
- [J28] **Let There be Direction in Hypergraph Neural Networks**, with S. Fiorini, M. Ciavotta, A. Del Bue, Transactions on Machine Learning Research (2024) [Publisher](#)
- [J27] **Using artificial intelligence and deep learning to optimise the selection of adult congenital heart disease patients in S-ICD screening**, with ElRefai, M., Abouelasaad, M., Conibear, I., Wiles, B.M., Dunn, A.J., Zemkoho, A.B., Morgan, J. and Roberts, P.R., Indian Pacing and Electrophysiology Journal 24 (4), 192-199 (2024) [Publisher](#)
- [J26] **A numerically exact algorithm for the bin-packing problem**, with R. Baldacci, J.F. Cordeau, F. Furini, INFORMS Journal on Computing, 36(1), pp. 141-162 (2024) [Publisher](#)
- [J25] **Deep learning and hyperparameter optimization for assessing one’s eligibility for a subcutaneous implantable cardioverter-defibrillator**, with A.J. Dunn, M. ElRefai, P.R. Roberts, B.M. Wiles, A.B. Zemkoho, Annals of Operations Research, 328(1):309–335 (2023) [Publisher](#) · [Optimization Online](#)
- [J24] **Correlation analysis of deep learning methods in S-ICD screening**, with M. ElRefai, M. Abouelasaad, B.M. Wiles, A.J. Dunn, A.B. Zemkoho, J.M. Morgan, P.R. Roberts, Annals of Noninvasive Electrocardiology, 2023, 28(4) [Publisher](#)
- [J23] **Role of deep learning methods in screening for subcutaneous implantable cardioverter defibrillator in heart failure**, with M. ElRefai, M. Abouelasaad, B.M. Wiles, A.J. Dunn, A.B. Zemkoho, J.M. Morgan, P.R. Roberts, Annals of Noninvasive Electrocardiology, 2023, 28(1), e13028 [Publisher](#)
- [J22] **Energy Storage Operation and Electricity Market Design: On the Market Power of Monopolistic Storage Operators**, with E. Bjorndal, M. Bjorndal, M.-F. Körner, C. Leinauer, M. Weibelzahl, European Journal of Operational Research, 307(2), 887–909 (2023) [Publisher](#)

- [J21] **Submodular maximization of concave utility functions combined with a set-union operator with applications to maximal covering location problems**, with F. Furini, I. Ljubic, *Mathematical Programming*, 195, 9–56 (2022) [Publisher](#) · [Optimization Online](#)
- [J20] **Deep learning-based insights on T:R ratio behaviour during prolonged screening for S-ICD eligibility**, with M. ElRefai, M. Abouelasaad, Benedict M. Wiles, Anthony J. Dunn, Alain B. Zemkoho, Paul R. Roberts, *Journal of Interventional Cardiac Electrophysiology* (first online 13th May 2022) [Publisher](#)
- [J19] **Deep learning methods for screening patients’ S-ICD implantation eligibility**, with A. Dunn, M. ElRefai, P. Roberts, B. Wiles, A. Zemkoho, *Artificial Intelligence in Medicine* 119 (2021) 102139 [Publisher](#) · [arXiv](#)
- [J18] **On the exact separation of cover inequalities of maximum depth**, with D. Catanzaro, F. Furini, *Optimization Letters* 16, 449–469 (2022) [Publisher](#) · [Optimization Online](#)
- [J17] **Optimizing over the Closure of Rank Inequalities with a Small Right-Hand Side for the Maximum Stable Set problem via Bilevel Programming**, with S. Gualandi, *INFORMS Journal on Computing* 34(2):1006–1023 (2021) [Publisher](#)
- [J16] **Airport Capacity Extension, Fleet Investment, and Optimal Aircraft Scheduling in a Multi-Level Market Model: Quantifying the Costs of Imperfect Markets**, with M. Sirvent, M. Weibelzahl, *OR Spectrum* 43 (2), 367–408 (2021) [Publisher](#)
- [J15] **A new combinatorial branch-and-bound algorithm for the Knapsack Problem with Conflicts**, with F. Furini, P. San Segundo, *European Journal of Operational Research*, 289 (2), 435–455 (2021) [Publisher](#) · [Optimization Online](#)
- [J14] **Elastic traffic engineering subject to a fair bandwidth assignment via bilevel programming**, with L. Gianoli, E. Amaldi, A. Capone, *IEEE/ACM Transactions on Networking*, 28 (6), 2407–2420 (2020) [Publisher](#)
- [J13] **A lexicographic pricer for the fractional bin packing problem**, with F. D’Andreagiovanni, F. Furini, *Operations Research Letters*, 47(6), 622–628 (2019) [Publisher](#)
- [J12] **Computing a Pessimistic Leader-Follower Equilibrium with Multiple Followers: the Mixed-Pure Case**, with A. Marchesi, N. Gatti, *Algorithmica*, 82(5), 1189–1238 (2019) [Publisher](#) · [arXiv](#)
- [J11] **Leadership in Singleton Congestion Games: What is Hard and What is Easy**, with A. Marchesi, M. Castiglioni, N. Gatti, *Artificial Intelligence (AIJ)*: 277, 1–31 (2019) [Publisher](#) · [arXiv](#)
- [J10] **Bilevel programming methods for computing single-leader-multi-follower equilibria in normal-form and polymatrix games**, with N. Basilico, A. Marchesi, N. Gatti, *EURO Journal on Computational Optimization*, 8(1), 3–31 (2019) [Publisher](#)
- [J9] **A new branch-and-bound algorithm for the maximum edge-weighted clique problem**, with P. San Segundo, F. Furini, I. Ljubic, *European Journal of Operational Research (EJOR)*: 278 (1), 76–90 (2019) [Publisher](#)
- [J8] **Lot sizing with storage losses under demand uncertainty**, with A. Koster, N. Spieckermann, *Journal of Combinatorial Optimization* 36: 763–788 (2018) [Publisher](#)
- [J7] **A workload-dependent task assignment policy for crowd-sourcing**, with I. Catallo, P. Fraternali, D. Martinenghi, *World Wide Web Journal*: 20 (6): 1179–1210 (2017) [Publisher](#)
- [J6] **Discrete optimization methods to fit piecewise-affine models to data**, with E. Amaldi, L. Taccari, *Computers and Operations Research*, 75: 214–230 (2016) [Publisher](#)
- [J5] **Data Uncertainty in Virtual Network Embedding: Robust Optimization and Protection Levels**, with A. Koster, M. Tieves, *Journal of Network and Systems Management*, 24(3): 681–710 (2016) [Publisher](#)
- [J4] **Coordinated cutting plane generation via multi-objective separation**, with E. Amaldi, S. Gualandi, *Mathematical Programming, series A*, 143(1–2): 87–110 (2014) [Publisher](#) · [Optimization Online](#)
- [J3] **Network optimization problems subject to max-min fair flow allocation**, with E. Amaldi, A. Capone, L. Gianoli, *IEEE Communications Letters*, 17(7): 1463–1466 (2013) [Publisher](#)
- [J2] **A distance-based point reassignment heuristic for the  $k$ -hyperplane clustering problem**, with E. Amaldi, *European Journal of Operations Research*, 227(1): 22–29 (2013) [Publisher](#)
- [J1] **On coordinated cutting plane generation and mixed integer programs with nonconvex 2-norm constraints**, *single author*, *4OR: A Quarterly Journal of Operations Research*, 11(1): 95–96 (2013) [Publisher](#)

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Peer-reviewed papers published in conference proceedings



- [P30] **Solving the 2-norm k-hyperplane clustering problem via multi-norm formulations**, *single author*, International Conference on Learning Representations (ICLR), Rio, Brazil, 2026 [OpenReview](#)
- [P29] **Directional Sheaf Hypergraph Networks: Unifying Learning on Directed and Undirected Hypergraphs**, with E. Mule, S. Fiorini, A. Purificato, F. Siciliano, F. Silvestri, International Conference on Learning Representations (ICLR), Rio, Brazil, 2026 [Publisher](#) · [arXiv](#)
- [P28] **Sheaves Reloaded: A Direction Awakening**, with S. Fiorini, H. Aktas, I. Duta, P. Morerio, A. Del Bue, P. Liò, International Conference on Learning Representations (ICLR), Rio, Brazil, 2026 [Publisher](#) · [arXiv](#)
- [P27] **Bilevel Optimization Model and a Heuristic Solution Method for Transformer Rating Optimization**, with F. Garuba, A. Martinez-Sykora, C. Tomasi, A.B. Zemkoho, Optimization and Decision Science (ODS), Milan, Italy, 2025  
Springer Proceedings in Mathematics and Statistics, to appear.
- [P26] **Graph Learning in 4D: a Quaternion-valued Laplacian to Enhance Spectral GCNs**, with S. Fiorini, M. Ciavotta, E. Messina, AAAI Conference on Artificial Intelligence AAAI, Vancouver, CA, 2024 [Publisher](#) · [arXiv](#)  
Proceedings of the 38th AAAI
- [P25] **SigMaNet: One Laplacian to Rule Them All**, with S. Fiorini, M. Ciavotta, E. Messina, AAAI Conference on Artificial Intelligence AAAI, Washington, US, 2023 [Publisher](#) · [arXiv](#)  
Proceedings of the 37th AAAI
- [P24] **Private Bayesian Persuasion with Sequential Games**, with N. Celli, N. Gatti, AAAI Conference on Artificial Intelligence AAAI, New York, US, 2020 [Publisher](#) · [arXiv](#)  
Proceedings of 37th AAAI
- [P23] **Computing Optimal Ex Ante Correlated Equilibria in Two-Player Sequential Games**, with A. Celli, N. Gatti, International Conference on Autonomous Agents and Multiagent Systems (AAMAS), Montreal, Canada, 2019 [Publisher](#)  
Proceedings of the 18th AAMAS
- [P22] **Leadership in Singleton Congestion Games**, with A. Marchesi, N. Gatti, International Joint Conference on Artificial Intelligence (IJCAI), Stockholm, Sweden, 2018 [Publisher](#)  
Proceedings of the 27th IJCAI, pages: 447–453
- [P21] **Facility location with item storage and delivery**, with J. Fliege, R. Walton, Optimization and Decision Science (ODS), Sorrento, Italy, 2017 [Publisher](#)  
Springer Proceedings in Mathematics and Statistics, A. Sforza, C. Sterle, eds., 2017
- [P20] **On the separation of topology-free rank inequalities for the max stable set problem**, with S. Gualandi, International Symposium on Experimental Algorithms (SEA), London, UK, 2017 [Publisher](#)  
Leibniz International Proceedings in Informatics, pages 29:1–29:13, 2017
- [P19] **Pessimistic Leader-Follower Equilibria with Multiple Followers**, with N. Gatti, A. Marchesi, International Joint Conference on Artificial Intelligence (IJCAI), Melbourne, Australia, 2017 [Publisher](#)  
Proceedings of the 26th IJCAI, pages: 171–177
- [P18] **Bilevel programming approaches to the computation of optimistic and pessimistic leader-multi-follower equilibria**, with N. Basilico, N. Gatti, A. Marchesi, International Symposium on Experimental Algorithms (SEA), London, UK, 2017 [Publisher](#)  
Leibniz International Proceedings in Informatics, pages 31:1–31:14, 2017
- [P17] **On robust lot sizing problems with storage deterioration, with applications to heat and power cogeneration**, with A. Koster, N. Spieckermann, International Symposium on Combinatorial Optimization (ISCO), Vietri Sul Mare, Salerno, Italy, 2016 [Publisher](#)  
Combinatorial Optimization, LNCS, 9849: 26–37, 2016
- [P16] **Methods for Finding Leader-Follower Equilibria with Multiple Followers**, with N. Basilico, N. Gatti, International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2016), Singapore, 2016 [Publisher](#) · [arXiv](#)
- [P15] **On the generation of cutting planes which maximize the bound improvement**, with M. Tieves, International Symposium on Experimental Algorithms (SEA), Paris, France, 2015 [Publisher](#) · [Optimization Online](#)  
Experimental Algorithms, LNCS, 9125: 97–109, 2015
- [P14] **Virtual network embedding under uncertainty: exact and heuristic approaches**, with A. Koster, M. Tieves, International Conference on Design of Reliable Communication Networks (DRCN), Kansas City, US, Missouri, 2015 [Publisher](#)  
IEEE Xplore, 10.1109/DRCN.2015.7148978, 1–8, 2015

- [P13] **On the computational complexity of the virtual network embedding problem**, with A. Koster, M. Tieves, International Network Optimization Conference (INOC), Warsaw, Poland, 2015 [Publisher](#)  
Electronic Notes in Discrete Mathematics, 52: 213–220, 2016
- [P12] **Maximum throughput network routing subject to fair flow allocation**, with E. Amaldi, L. Taccari, International Symposium on Combinatorial Optimization (ISCO), Lisbon, Portugal, 2014 [Publisher](#)  
Combinatorial Optimization, LNCS, 8596: 1–12, 2014.
- [P11] **Bound-optimal cutting planes**, single author, Cologne-Twente Workshop on Graphs and Combinatorial Optimization (CTW), Enschede, Netherlands, 2013 [Proceedings](#)
- [P10] **Energy-aware traffic engineering with elastic demands and MMF bandwidth allocation**, with E. Amaldi, A. Capone, L. Gianoli, IEEE International Workshop on Computer Aided Modeling and Design of Communication Links and Networks (CAMAD), IEEE, Berlin, Germany, 2013 [Publisher](#)  
IEEE Xplore, 10.1109/CAMAD.2013.6708111, 169–174, 2013
- [P9] **On single-path network routing subject to max-min fair flow allocation**, with E. Amaldi, L. Gianoli, C. Ileri, International Network Optimization Conference (INOC), Tenerife, Spain, 2013 [Publisher](#)  
Electronic Notes in Discrete Mathematics, 41: 543–550, 2013
- [P8]  **$k$ -Piecewise Affine Model Fitting: Heuristics based on multiway linear classification**, with E. Amaldi, L. Taccari, Cologne-Twente Workshop on Graphs and Combinatorial Optimization (CTW), Munich, Germany, 2012
- [P7] **The impact of the norm on the  $k$ -Hyperplane Clustering problem: relaxations, restrictions, approximation factors, and exact formulations**, single author, Cologne-Twente Workshop on Graphs and Combinatorial Optimization (CTW), Frascati (Rome), Italy, 2011
- [P6] **Formulations and heuristics for the  $k$ -Piecewise Affine Model Fitting problem**, with E. Amaldi, L. Taccari, Cologne-Twente Workshop on Graphs and Combinatorial Optimization (CTW), Frascati (Rome) Italy, 2011
- [P5] **Improving cutting plane generation with 0-1 inequalities by bi-criteria separation**, with E. Amaldi, S. Gualandi, International Symposium on Experimental Algorithms (SEA), Ischia (Naples), Italy, 2010 [Publisher](#)  
Experimental Algorithms, LNCS, 6049: 266–275, 2010
- [P4] **Single and multi-group evacuation in medium and maxi health-care emergencies**, with E. Amaldi, C. Iuliano, Operations Research Applied to Health Services Conference (ORAHS), Genoa, Italy, 2010
- [P3] **Optimization models for injured people evacuation in medium/maxi health-care emergencies**, with E. Amaldi, C. Iuliano, Workshop on Health Care Management (WHCM), IEEE, Venice, Italy, 2010 [Publisher](#)  
IEEE Xplore, 10.1109/WHCM.2010.5441250, 1–6, 2010
- [P2]  **$k$ -hyperplane clustering problem: column generation and a metaheuristic**, with E. Amaldi, K. Dhyan, Cologne-Twente Workshop on Graphs and Combinatorial Optimization (CTW), Paris, France, 2009
- [P1] **An adaptive point-reassignment metaheuristic for the  $k$ -hyperplane clustering problem**, with E. Amaldi, Metaheuristic International Conference (MIC), Hamburg, Germany, 2009

## Invited talks and seminars

- [M22] **Advances in Spectral Graph and Hypergraph Learning: towards more expressive Laplacian matrices**, Invited by Prof. S. Gualandi, Università di Pavia, Pavia, 2025
- [M21] **Graph and Hypergraph Learning via Complex- and Quaternion-Valued Spectral Convolutional Operators**, MATH4AI, UMI Workshop, Bari, Italy, 2025  
Keynote invited by UMI Group Mathematics for Artificial Intelligence and Machine Learning
- [M20] **Integer programming and norm minimization (in machine learning?)**, Invited by Prof. P. Avella, Università del Sannio, Benevento, 2023
- [M19] **Bound-optimal cutting-plane generation**, Invited by G. Eichfelder, J. Kronqvist, A. Lodi, F. Oliveira, E. Rönnberg, Learning from Both Sides Linear and Nonlinear Mixed-Integer Optimization @ Institute Mittag-Leffler, Djursholm (SWE), 2023
- [M18] **A Numerically-Exact Algorithm for the Bin Packing Problem**, Invited by Prof. L. Mariani, Università di Milano-Bicocca, Milan, 2023
- [M17] **Minimizing vector norms with integer programming**, Invited by Prof. S. Gualandi, Università di Pavia, Pavia (virtually), 2022
- [M16] **Integer programming and norm minimization for data science**, Invited by Prof. E. Messina, Università di Milano-Bicocca, Milan (virtually), 2021

- [M15] **Bilevel programming and the computation of pessimistic single-leader-multi-follower equilibria in Stackelberg games**, *Invited by Dr. S. Toubaline*, LAMSADE, Université Paris-Dauphine, Paris, FR, 2019
- [M14] **The bin packing problem with item class setups**, *Invited by Prof. E. Amaldi*, Politecnico di Milano, Department of Electronics, Information, and Biotechnologies, Milan, 2019
- [M13] **Bilevel programming and the computation of pessimistic single-leader-multi-follower equilibria in Stackelberg games**, *Invited by Dr. T. Oertel*, Cardiff University, Department of Mathematics, Cardiff, 2019
- [M12] **Computing Optimistic and Pessimistic Leader-Follower Equilibria with Multiple Followers**, *Invited by Dr. R. Misener*, Imperial College London, Department of Computing, London, 2018
- [M11] **Network routing through the Internet as a Stackelberg game**, *Invited by Prof. T. Ralphs*, Lehigh University, Department of Industrial and Systems Engineering, Bethlehem, US, Pennsylvania, 2018
- [M10] **Exact single level and bilevel programming approaches to the computation of Leader-Follower Nash Equilibria in games with many followers**, *Invited by Dr. L. Tran-Thanh*, University of Southampton, AIC (Agents, Interactions and Complexity) seminar series, Department of Electronics and Computer Science, Southampton, 2017
- [M9] **Network routing through the Internet as a Stackelberg game**, *Invited by Prof. H. Xu*, University of Southampton, Business Advisory Board meeting, CORMSIS (Centre for Operational Research, Management Science and Information Systems), University of Southampton, 2017
- [M8] **Exact single level and bilevel programming approaches to the computation of Leader-Follower Nash Equilibria in games with many followers**, *Invited by Dr. S. Gualandi*, Pavia University, Pavia, 2017
- [M7] **Network routing through the Internet as a Stackelberg game**, *Invited by Prof. A. Koster*, RWTH Aachen University, Aachen, 2017
- [M6] **Computing Leader-Follower Nash Equilibria in games with many followers via optimistic and pessimistic bilevel programming**, *Invited by Prof. A. Koster*, RWTH Aachen University, Aachen, 2016
- [M5] **On bilevel programming and the computation of Leader-Follower (Nash) Equilibria in games with many followers**, *Invited by Prof. N. Gatti*, Politecnico di Milano, Milan, 2016
- [M4] **(Bilevel programming and) bound-optimal cutting plane generation**, *Invited by Prof. T. Ralphs*, Department of Industrial and Systems Engineering, Lehigh University, Bethlehem, US, Pennsylvania, 2015
- [M3] **On the exact separation of rank inequalities for the maximum stable set problem**, *Invited by Dr. F. D'Andreagiovanni*, Zuse Institute Berlin (ZIB), Berlin, Germany, 2014
- [M2] **Coordinated cutting plane generation via multi-objective separation and extensions**, *Invited by Prof. A. Koster*, RWTH Aachen University, Aachen, Germany, 2012
- [M1] **Applications of exact mathematical programming**, *Invited by Dr. M. Pozzi*, Centro Elettronica Sperimentale Italiano, Italian Center for Experimental Electronics (CESI), Milan, Italy, 2011

## Conference oral/poster presentations

- [O38+M22] **Graph and Hypergraph Learning via Complex- and Quaternion-Valued Spectral Convolutional Operators**, *3rd Workshop of UMI Group Mathematics for Artificial Intelligence and Machine Learning (MATH4AI 2025)*, Bari, Italy, 2025  
Keynote invited by UMI Group Mathematics for Artificial Intelligence and Machine Learning
- [O37] **Graph and Hypergraph Learning via Complex- and Quaternion-Valued Spectral Convolutional Operators**, *Economics Winter Seminar 2025*, Castione della Presolana, Italy, 2025
- [O38] **Graph Learning in 4D: a Quaternion-valued Laplacian to Enhance Spectral GCNs**, *AAAI Conference on Artificial Intelligence (AAAI 2024)*, Vancouver, Canada
- [O37] **Investment models in the energy and aviation markets via multi-level optimization**, *Economics Winter Seminar 2024*, Castione della Presolana, Italy, 2024
- [O36+M19] **Bound-optimal cutting-plane generation**, *with T. Ralphs*, *Learning from Both Sides Linear and Nonlinear Mixed-Integer Optimization @ Institute Mittag-Leffler*, Djursholm (SWE), 2023  
Invited by G. Eichfelder, J. Kronqvist, A. Lodi, F. Oliveira, E. Rönnberg
- [O35] **Mixed integer bilevel programming methods for solving pessimistic multi-follower bilevel problems over simplices.**, *1st International Conference on Bilevel Optimization (ICBO2023)*, 2023  
Southampton, UK

- [O34] **A Numerically-Exact Algorithm for the Bin Packing Problem**, with *R. Baldacci, F. Cordeau, F. Furini*, 31th Annual conference of the Association of European Operational Research Societies (EURO), Espoo, Finland, 2022
- [O33] **Norm minimization problems in data science: an integer programming perspective**, ECCO XXXV - CO 2022 Joint Conference, Online, 2022
- [O32] **Infinite-Precision Pattern-Enumeration Techniques for Bin Packing Problems**, with *R. Baldacci, F. Furini*, 63th Annual Meeting of the Operational Research Society (OR63), Online, 2021
- [O31] **Decomposition methods for maximizing a submodular function combined with a set-union operator**, with *F. Furini, I. Ljubic*, 3rd IMA/OR Society Conference, Online, 2021
- [O30] **Decomposition methods for maximizing a submodular function combined with a set-union operator**, with *F. Furini, I. Ljubic*, International Conference on Optimization and Decision Science (ODS), Online, 2020
- [O29] **The bin packing problem with item class setups**, with *F. Furini, I. Ljubic*, 30th Annual conference of the Association of European Operational Research Societies (EURO), Dublin, Ireland, 2019
- [O28] **Computing Pessimistic Leader-Follower Equilibria with Multiple Followers (in the mixed-pure case)**, with *N. Gatti, A. Marchesi*, International Symposium on Mathematical Programming (ISMP), Bordeaux, France, 2018
- [O27] **Computing Leader-follower equilibria with multiple followers under the assumption of pessimism in the mixed-pure case**, with *N. Gatti, A. Marchesi*, International Workshop on Bilevel Programming (IWOBP), Lille, France, 2018
- [O26] **Using Mathematical Optimisation and Game Theory to improve the speed of broadband connections over the Internet**, with *E. Amaldi, A. Capone, L. Gianoli*, STEM for Britain, House of Commons (Westminster), London, UK, 2017
- [O25] **On the separation of topology-free rank inequalities for the max stable set problem**, with *S. Gualandi*, International Symposium on Experimental Algorithms (SEA), London, UK, 2017  
Leibniz International Proceedings in Informatics, pages 21:1–29:13, 2017
- [O24] **Bilevel programming approaches to the computation of optimistic and pessimistic leader-multi-follower equilibria**, with *N. Basilico, N. Gatti, A. Marchesi*, International Symposium on Experimental Algorithms (SEA), London, UK, 2017  
Leibniz International Proceedings in Informatics, pages 31:1–31:14, 2017
- [O23] **On the computational complexity of the virtual network embedding problem**, with *A. Koster, M. Tieves*, International Network Optimization Conference (INOC), Warsaw, Poland, 2015  
Electronic Notes in Discrete Mathematics, 52: 213–220, 2016
- [O22] **On the generation of cutting planes which maximize the bound improvement**, with *M. Tieves*, International Symposium on Experimental Algorithms (SEA), Paris, France, 2015  
Experimental Algorithms, LNCS, 9125: 97–109, 2015
- [O21] **Dynamic robust optimization approaches to lot sizing**, with *A. Koster, N. Spiekermann, L. Taccari*, Modeling and optimization: theory and applications (MOPTA), Bethlehem, US, Pennsylvania, 2015
- [O20] **Applications of bilevel programming to cutting plane generation**, with *M. Tieves, S. Gualandi*, International Symposium on Mathematical Programming (ISMP), Pittsburgh, US, Pennsylvania, 2015
- [O19] **Bound optimal cutting planes: on the generation of cuts which maximize the bound improvement**, Annual Conference of the French Operations Research Society (ROADEF), Marseille, FR, 2015
- [O18] **On the exact separation of rank inequalities for the maximum stable set problem**, with *S. Gualandi*, Integer Programming and Combinatorial Optimization (IPCO), Bonn, Germany, 2014
- [O17] **Maximum throughput network routing subject to fair flow allocation**, with *E. Amaldi, L. Taccari*, International Symposium on Combinatorial Optimization (ISCO), Lisbon, Portugal, 2014  
Combinatorial Optimization, LNCS, 8596: 1–12, 2014.
- [O16] **Virtual network embedding with network design elements: reservation of physical resources**, with *A. Koster, M. Tieves*, INFORMS Annual Meeting, San Francisco, US, California, 2014
- [O15] **On the generation of cuts which maximize the bound improvement**, with *M. Tieves*, Annual Conference of the Italian Operational Research Society (AIRO), Como, Italy, 2014  
Within the “Mixed-integer linear programming” organized session, chaired by myself and S. Gualandi.
- [O14] **On the exact separation of rank inequalities for the stable set problem**, with *S. Gualandi*, International Symposium on Combinatorial Optimization (ISCO), Lisbon, Portugal, 2014



- [O13] **Bilevel optimization models for traffic engineering with elastic demands and fair flow allocation**, with E. Amaldi, A. Capone, L. Gianoli, INFORMS Telecommunications Conference (INFORMS TELECOM), Lisbon, Portugal, 2014
- [O12] **Bound-optimal cutting planes**, *single author*, Cologne-Twente Workshop on Graphs and Combinatorial Optimization (CTW), Enschede, Netherlands, 2013
- [O11] **Valid inequalities and lifting for the 0-1 Gamma-robust knapsack problem**, with C. Büsing, A. Koster, European Conference on Operational Research (EURO), Rome, Italy, 2013
- [O39] **Spatial branch-and-bound methods for solving the  $k$ -Hyperplane clustering**, with J. Montree, 22nd EUROPT Conference on Advances in Continuous Optimization (EUROPT 2025), Southampton, UK, 2025  
Work subsequently published at ICLR 2026 as “Solving the 2-norm  $k$ -hyperplane clustering problem via multi-norm formulations”
- [O10] **Spatial branch-and-bound for nonconvex Euclidean norm constrained mathematical programs**, with F. Margot, International Symposium on Mathematical Programming (ISMP), Berlin, Germany, 2012
- [O9] **The impact of the norm on the  $k$ -Hyperplane Clustering problem: relaxations, restrictions, approximation factors, and exact formulations**, *single author*, Cologne-Twente Workshop on Graphs and Combinatorial Optimization (CTW), Frascati (Rome), Italy, 2011
- [O8] **Multi-objective separation for sequentially coordinated cutting plane generation**, with E. Amaldi, S. Gualandi, Combinatorial Optimization Workshop (COW), Aussois, France, 2011
- [O7] **(Beyond) sequentially coordinated cutting plane generation for inequalities with integer coefficients**, with E. Amaldi, S. Gualandi, Mixed Integer Programming Workshop (MIP), Waterloo, Canada, Ontario, 2011
- [O6] **Improving cutting plane generation with 0-1 inequalities by bi-criteria separation**, with E. Amaldi, S. Gualandi, International Symposium on Experimental Algorithms (SEA), Ischia (Naples), Italy, 2010  
Experimental Algorithms, LNCS, 6049: 266–275, 2010
- [O5] **Exact bi-criteria cutting plane generation for inequalities with 0-1 coefficients**, with E. Amaldi, S. Gualandi, Mixed Integer Programming Workshop (MIP), Atlanta, US, Georgia, 2010
- [O4] **An adaptive point-reassignment metaheuristic for the  $k$ -hyperplane clustering problem**, with E. Amaldi, Metaheuristic International Conference (MIC), Hamburg, Germany, 2009
- [O3]  **$k$ -Hyperplane Clustering: an adaptive point-reassignment algorithm**, with E. Amaldi, International Symposium on Mathematical Programming (ISMP), Chicago, US, Illinois, 2009
- [O2] **Towards a decision support system for injured people evacuation in medium/maxi emergencies**, with E. Amaldi, C. Iuliano, PhD Alumni Association Meeting (PhDei), Milan, Italy, 2009
- [O1] **Discrete optimization algorithms for the  $k$ -hyperplane clustering problem**, with E. Amaldi, K. Dhyani, Annual Conference of the Italian Operational Research Society (AIRO), Ischia (Naples), Italy, 2008

## Activities as conference organizer

- [CO6] **HEXAGON2025**, *The second HEXAGON Workshop on Power Grids*, Benevento, Italy, September 2025
- [CO6] **HEXAGON2024**, *The HEXAGON Workshop on Power Grids*, Bergamo, Italy, June 2024
- [CO5] **ICBO2023**, *1st International Conference on Bilevel Optimization*, Southampton, UK, August 2023
- [CO4] **OR63**, *63rd Annual conference of the Operational Research Society*, University of Southampton (Virtual), September 2021
- [CO3] **W-GREENBELT 2018**, *Workshop on GREEN and BluE Logistics and Transport*, University of Southampton, 16-17 July 2018
- [CO2] **ICCL 2017**, *International Conference on Computational Logistics*, University of Southampton, 18-20 October 2017
- [CO1] **BODAF 2017**, *(Bilevel) Optimization, Data Analysis and Forecasting*, University of Southampton, 3-4 July 2017

## Activities as conference stream organizer

- [CStO2] **64th Annual Conference of the Operational Research Society 2022**, (Warwick, UK), *stream title: “Combinatorial Optimisation”*

- [CStO1] **63rd Annual Conference of the Operational Research Society 2021**, (*Online, UK*), stream title: “Combinatorial Optimisation”; sessions: “Integer programming and combinatorial optimisation”, “Heuristic and scheduling problems”, “Submodularity, robustness and nonlinear optimisation”, speakers: I. Aliev, S. Gualandi, Y. Lu, V.T. Ramamoorthy, D. Paraskevopoulos, C. Lamas-Fernandez, J. Ward, F. Garuba, M. Anjos, and myself

## Activities as conference session organizer

- [CSO6] **ICBO2023**, *1st International Conference on Bilevel Optimization*, Southampton, UK, session title: “(Mixed) integer (nonlinear) bilevel optimization”, jointly organized with M. Schmidt
- [CSO5] **IMA and OR Society Conference on Mathematics of Operational Research 2021**, (*Online, UK*), session title: “(Multilevel) optimization methods for energy transmission”, speakers: L. Schewe, M. Weibelzahl, P. Bartmeyer
- [CSO4] **European Conference on Operational Research (EURO) 2019**, (*Dublin, IR*), session title: “Packing and Covering with Integer Programming”, speakers: P. Fath, F. Antonio Medrano, S. Coniglio, M. Sorensen
- [CSO3] **IMA and OR Society Conference on Mathematics of Operational Research 2017**, (*Birmingham, UK*), session title: “Nonlinear and Bilevel Optimisation”, speakers: M. Sinnl, A. Zemkoho, S. Brierley
- [CSO2] **International Symposium on Mathematical Programming (ISMP) 2015**, (*Pittsburgh, US, Pennsylvania*), session title: “Applications of Bilevel Programming to Combinatorial Optimization and Game Theory”, speakers: M. Carvalho, L. Lozano, C. Casorran-Amilburu
- [CSO1] **Annual Conference of the Italian Operations Research Society (AIRO) 2014**, (*Como, Italy*), session title: “Mixed-integer linear programming”, speakers: S. Smriglio, M. Boccia, S. Coniglio

## Visits for scientific collaborations

- [I12] **Università La Sapienza, Rome**, February 2023, invited by Prof. F. Furini
- [I11] **Dauphine University, Paris**, November 2019, invited by Dr. F. Furini
- [I10] **Dauphine University, Paris**, June/July 2019, invited by Dr. F. Furini
- [I9] **Dauphine University, Paris**, October 2018, invited by Dr. F. Furini
- [I8] **Dauphine University, Paris**, May 2018, invited by Dr. F. Furini
- [I7] **Lehigh University, Bethlehem**, April 2018, invited by Prof. T. Ralphs
- [I6] **RWTH Aachen University, Aachen**, March 2017, invited by Prof. A. Koster
- [I5] **RWTH Aachen University, Aachen**, April 2016, invited by Prof. A. Koster
- [I4] **Politecnico di Milano, Milan**, March 2016, invited by Prof. N. Gatti
- [I3] **Lehigh University, Bethlehem**, July 2015, invited by Prof. T. Ralphs
- [I2] **Zuse Institute Berlin (ZIB), Berlin**, December 2014, invited by Dr. F. D’Andreagiovanni
- [I1] **Politecnico di Milano, Milan**, a number of times since September 2012, invited by Prof. E. Amaldi and N. Gatti
- [I0] **Carnegie-Mellon University, Pittsburgh**, January-March 2010, invited by Prof. F. Margot

## Researchers invited for scientific collaborations

- [V5] **S. Gualandi (University of Pavia)**, November 2018, invited to University of Southampton
- [V4] **F. Furini (Dauphine University)**, August 2018, invited to University of Southampton
- [V3] **T. Ralphs (Lehigh University)**, April 2017, invited to University of Southampton
- [V2] **L. Taccari (Politecnico di Milano)**, March 2015, invited to RWTH Aachen University
- [V1] **S. Gualandi (AntOptima)**, November 2013, invited to RWTH Aachen University

## Activities as editorial board member

2020 - **Associate Editor**, *Asia-Pacific Journal of Operational Research*

## Activities as reviewer

- National research evaluation **VQR 2015-19 (Italian Research quality evaluation)**, Referee, ERC codes: PE1\_15 (Discrete mathematics and combinatorics), PE6\_6 (Algorithms, distributed, parallel and network algorithms, algorithmic game theory), PE6\_7 (Artificial intelligence, intelligent systems, multi agent systems), PE6\_11 (Machine learning, statistical data processing and applications using signal processing (e.g., speech, image, video))
- Grants **Chilean National Fund for Scientific and Technological Development, FONDECYT (FONDECYT-CHILE)**, 2022

- Journals
- 4OR**
  - Annals of OR**
  - Applied Soft Computing**
  - Computational and Applied Mathematics**
  - Computer Networks**
  - Computers and Operations Research**
  - Discrete Optimization**
  - Energy**
  - EURO Journal on Computational Optimization**
  - European Journal of Operational Research**
  - European Journal on Computational Optimization**
  - Expert Systems with Applications**
  - IIIE Transactions**
  - IEEE Transactions on Cloud Computing**
  - IEEE Transactions on Network and Service Management**
  - INFOR (Information Systems and Operational Research)**
  - INFORMS Networks**
  - INFORMS Journal on Computing**
  - INFORMS Operations Research**
  - International Journal of Hydrogen Energy**
  - International Journal of Parallel, Emergent and Distributed Systems**
  - Journal of Experimental Algorithmics**
  - Journal of the Operational Research Society**
  - Operations Research Letters**
  - Optimization Letters**
  - RAIRO Operations Research**
  - SIAM Journal of Optimization**
  - TOP (Journal of the Spanish Society of Statistics and Operations Research)**
- Conferences
- NeurIPS 2025**, *Annual Conference on Neural Information Processing Systems*
  - ODS 2025**, *Optimization and Data Science*
  - ECAI 2025**, *European Conference on Artificial Intelligence*
  - AAAI 2025**, *AAAI Conference on Artificial Intelligence*
  - EvoStar 2024**
  - AAAI 2024**, *AAAI Conference on Artificial Intelligence*
  - EvoApps 2023**
  - IPCO 2023**, *Conference on Integer Programming and Combinatorial Optimization*
  - EvoStar 2022**
  - WI 2021**, *Internationale Tagung Wirtschaftsinformatik (International Conference on Information Systems)*
  - STACS 2019**, *Symposium on Theoretical Aspects of Computer Science*
  - IJCAI 2018**, *International Joint Conference on Artificial Intelligence*
  - ICCL 2018**, *International Conference on Computational Logistics*
  - AAAI 2018**, *AAAI Conference on Artificial Intelligence*
  - ICCL 2017**, *International Conference on Computational Logistics*
  - ICAT 2017**, *International Conference on Information, Communication and automation technologies*
  - EvoCOMNET 2017**, “Application of Nature-inspired Techniques for Communication Networks and other Parallel and Distributed Systems” track within EvoSTAR 2017
  - AAAI 2017**, *AAAI Conference on Artificial Intelligence*
  - IJCAI 2016**, *International Joint Conference on Artificial Intelligence*
  - AAAI 2016**, *AAAI Conference on Artificial Intelligence*
  - CP 2015**, *Conference on Principles and Practice of Constraint Programming*
  - ENIC 2015**, *European Network Intelligence Conference*

**INOC 2015**, *International Network Optimization Conference*  
**OPODIS 2014**, *International Conference on Principles of Distributed Systems*  
**DRCN 2014**, *Design of Reliable Communication Networks*  
**AAAI 2014**, *AAAI Conference on Artificial Intelligence*  
**SEA 2013**, *Symposium on Experimental Algorithms*

## Activities as program committee member

**EvoApplications 2026**, *29th International Conference on the Applications of Evolutionary Computation*  
**AAAI 2026**, *AAAI Conference on Artificial Intelligence*  
**NeurIPS 2025**, *Annual Conference on Neural Information Processing Systems*  
**ECAI 2025**, *European Conference on Artificial Intelligence*  
**AAAI 2025**, *AAAI Conference on Artificial Intelligence*  
**AAAI 2024**, *AAAI Conference on Artificial Intelligence*  
**EvoAPPS 2024**, *27th International Conference on the Applications of Evolutionary Computation (part of EvoStar2024))*  
**AAAI 2023**, *AAAI Conference on Artificial Intelligence*  
**EvoAPPS 2023**, *26th International Conference on the Applications of Evolutionary Computation (part of EvoStar2023)*  
**EvoAPPS 2021**, *24th European Conference on the Applications of Evolutionary Computation, (part of EvoStar2021)*  
**EvoAPPS 2020**, *23rd European Conference on the Applications of Evolutionary and bio-inspired Computation, (part of EvoStar2020)*  
**IJCAI 2019**, *International Joint Conference on Artificial Intelligence*  
**ICCL 2018**, *International Conference on Computational Logistics*  
**IJCAI-ECAI 2018**, *International Joint Conference on Artificial Intelligence and European Conference on Artificial Intelligence*  
**AAAI 2018**, *AAAI Conference on Artificial Intelligence*  
**EvoCOMNET 2018**, *(workshop of EvoStar 2018)*  
**ICCL 2017**, *International Conference on Computational Logistics*  
**EvoCOMNET 2017**, *(workshop of EvoSTAR 2017)*  
**AAAI 2017**, *AAAI Conference on Artificial Intelligence*  
**IJCAI 2016**, *International Joint Conference on Artificial Intelligence*  
**AAAI 2016**, *AAAI Conference on Artificial Intelligence*

## Participation to conferences and workshops

### As an organizer

2024 **HEXAGON2024**, *The HEXAGON Workshop on Power Grids*, Bergamo, Italy, June 2023

### As an organizer & presenter

2023 **1st International Conference on Bilevel Optimization (ICBO2023)**, 2023, Southampton, UK  
2021 **63th Annual Meeting of the Operational Research Society (OR63)**, 2021, Online  
2019 **OptML: Optimization and Machine Learning**, 2019, Southampton, UK  
2018 **W-GREENBELT 2018, Workshop on GREEN and BluE Logistics and Transport**, 2018, Southampton, UK  
2017 **International Conference on Computational Logistics**, 2017, Southampton, UK  
· **(Bilevel) Optimization, Data Analysis and Forecasting**, 2017, Southampton, UK

### With an oral/poster presentation

**3rd Workshop of UMI Group Mathematics for Artificial Intelligence and Machine Learning**, Bari, Italy  
**Economics Winter Seminar 2025**, *Castione della Presolana, Italy*  
2024 **AAAI Conference on Artificial Intelligence (AAAI 2024)**, *Vancouver, Canada*  
**Economics Winter Seminar 2024**, *Castione della Presolana, Italy*

- 2023 **Learning from Both Sides (Linear and Nonlinear Mixed-Integer Optimization)**, 2022, Institute Mittag-Leffler, Djursholm, Sweden
- 2022 **33rd Annual conference of the Association of European Operational Research Societies (EURO 2022)**, 2022, Espoo, Finland
- 2021 **63th Annual Meeting of the Operational Research Society (OR63)**, 2021, Online
- . **3rd IMA/OR Society Conference**, 2021, Online
- 2020 **International Conference on Optimization and Decision Science (ODS)**, 2020, Online
- 2019 **30th Annual conference of the Association of European Operational Research Societies (EURO 2019)**, 2019, Dublin, Ireland
- 2018 **International Symposium on Mathematical Programming (ISMP)**, 2018, Bordeaux, France
- . **International Workshop on Bilevel Programming (IWOBIP)**, 2018, Lille, France
- 2017 **STEM for Britain, House of Commons (Westminster)**, 2017, London, UK
- Symposium on Experimental Algorithms (SEA)**, 2017, London, UK
- 2015 **Modeling and Optimization: Theory and Applications (MOPTA)**, 2015, Bethlehem, US, Pennsylvania
- . **International Symposium on Mathematical Programming (ISMP)**, 2015, Pittsburgh, US, Pennsylvania
  - . **Symposium on Experimental Algorithms (SEA)**, 2015, Paris, France
  - . **International Network Optimization Conference (INOC)**, 2015, Warsaw, Poland
  - . **Conference of the French Society of Operations Research and Decision Support (ROADEF)**, 2015, Marseille, France
- 2014 **Integer Programming and Combinatorial Optimization (IPCO)**, 2014, Bonn, Germany
- . **INFORMS Annual Meeting**, 2014, San Francisco, US, California
  - . **International Symposium on Combinatorial Optimization (ISCO)**, 2014, Lisbon, Portugal
  - . **INFORMS Telecommunications Conference**, 2014, Lisbon, Portugal
- 2013 **EURO-INFORMS Joint International Meeting**, 2013, Rome, Italy
- . **Cologne-Twente Workshop on Graphs and Combinatorial Optimization (CTW)**, 2013, Enschede, Netherlands
- 2012 **International Symposium on Mathematical Programming, ISMP**, 2012, Berlin, Germany
- 2011 **Cologne-Twente Workshop on Graphs and Combinatorial Optimization (CTW)**, 2011, Rome, Italy
- . **Mixed-Integer Programming Workshop (MIP)**, 2011, Waterloo, Canada, Ontario
  - . **Combinatorial Optimization Workshop (COW)**, 2011, Aussois, France
- 2010 **Symposium on Experimental Algorithms (SEA)**, 2010, Naples, Italy
- . **Mixed-Integer Programming Workshop (MIP)**, 2010, Atlanta, US, Georgia
- 2009 **International Symposium on Mathematical Programming, ISMP**, 2009, Chicago, US, Illinois
- 2008 **Annual Conference of the Italian Operations Research Society (AIRO)**, 2008, Naples, Italy
- . **Metaheuristics International Conference (MIC)**, 2008, Hamburg, Germany
- As a delegate
- 2025 **9th AIRO Young Workshop**, 2025, Pavia, IT
- I acted as member for the Fedegari Best Poster Prize committee
- 2024 **Four Motors for Europe – Artificial Intelligence as an opportunity for**, *Four Motors for Europe – presidency handover to Lombardy*, 2024, Stuttgart, DE
- 2024 **Mathematics for Artificial Intelligence and Machine Learning (MATH4AI 2023)**, 2024, Milan, Italy
- 2023 **Ultraoptymal Workshop**, 2023, Bergamo, Italy
- . **L’Ufficio per il Processo tra presente e futuro nell’Italia del nord-ovest**, 2023, Turin, Italy
- 2022 **DSA Summer conference**, 2022, Milan, Italy
- . **Workshop on Quantum Computing and Operations Research**, 2022, Fields Institute, Toronto (Online)
  - . **AI UK 2022**, *Alain Turing Institute*, 2022, ATI (Online)
- 2021 **Mixed Integer Programming Workshop**, 2021, Online



- 2020 **Mathematical Foundations of Optimisation in Data Science**, Newton Institute, 2020, Online
- . **ISCO (International Symposium on Combinatorial Optimization)**, 2020, Online
- 2016 **Annual Analytics Summit**, OR Society, 2016, London, UK
- . **First Aachen Discrete Mathematics Days**, 2016, Aachen, Germany
- . **Combinatorial Optimization Workshop (COW)**, 2016, Aussois, France
- 2015 **Mixed-Integer Nonlinear Methods for Energy Systems Engineering (MINOMESE)**, 2015, Aachen, Germany
- 2013 **Combinatorial Optimization Workshop (COW)**, 2013, Aussois, France
- 2009 **Integer Programming and Combinatorial Optimization (IPCO)**, 2009, Bertinoro, Italy

## Selected public engagement and AI training

- 2026 **Speaker, XXII ACBGroup Convention**, “*Etica della responsabilità, cultura economica e scenari geopolitici. Progettare il futuro tra sviluppo, impresa e contesto globale*”, Teatro Sociale, Bergamo, 25 September 2026  
Round table on the role of professionals in sustainable development and on technological transformation
- 2026 **Institutional opening**, “*Archeologie letterarie e artistiche dell’intelligenza artificiale*”, University of Bergamo, 2 October 2026  
Participation as Rector’s Delegate for Artificial Intelligence
- 2026 **Training session**, “*Come l’IA ha iniziato a parlare: una sguardo storico-scientifico sull’IA moderna*”, *Centro Scolastico La Traccia*, Calcinatè (BG), 3 September 2026  
Three-hour training session for teaching staff
- 2026 **University-wide training cycle on the use of artificial intelligence**, *Faculty and technical-administrative staff*, University of Bergamo  
Training on the use and adoption of modern AI and generative-AI tools
- 2026 **Panelist**, “*Institutional Policies and Guidelines on AI*”, *Tomorrow Today in Research Italia 2026* (Elsevier), Università Cattolica del Sacro Cuore, Milan, 11 June 2026  
Panel discussion on institutional AI governance and the conditions for responsible, trustworthy, and strategically aligned AI adoption
- 2026 **AI training seminar**, *Femca CISL Bergamo*, Bergamo  
Training session for council members
- 2026 **Invited speaker**, “*Intelligenza Artificiale e Diritto di Famiglia*”, *ONDIF Bergamo*, University of Bergamo  
Two-session accredited workshop on AI and law
- 2026 **Invited speaker**, “*Le nuove sfide dell’intelligenza artificiale. Diritto e Responsabilità*”, *SACBO / University of Bergamo*, Bergamo, 7 May 2026
- 2026 **Discussant, presentation of *Algomedia* (2026)**, with *Pietro Conte* and *Anna Caterina Dalmasso*, “*Prismi – Sei parole chiave per un’ecologia dell’intelligenza artificiale*”, University of Bergamo, 7 May 2026
- 2026 **Invited speaker**, “*Governare l’intelligenza artificiale. Riflessi di una rivoluzione su economia, lavoro, scuola e democrazia*”, *Fondazione Gritti Minetti / Officina Democratica*, Sala Galmozzi, Bergamo, 7 May 2026  
With Anna Ascani, Simone Varva, and Silvia Patrizia Gadda
- 2026 **Invited seminar on artificial intelligence**, *ITEMA AI Academy*, Itema Group, Colzate (BG)
- 2025 **Panelist**, “*Innovazione e sviluppo del territorio*”, *XX Congresso Territoriale CISL Bergamo*, Bergamo, 11 April 2025  
Contribution: “*Uno sguardo sull’IA moderna*”. With Giorgio Gori, Claudia Terzi, and Maria Paola Esposito; moderated by Silvana Galizzi

## Administrative service

- 2026– **Member of the Scientific and Technical Committee**, *RAISE – Research Centre for Responsible Artificial Intelligence, Society and Ethics*, University of Bergamo
- 2025– **Member of the doctoral board and curriculum coordinator**, *PhD in Artificial Intelligence for Sustainable Futures*, University of Bergamo  
Co-designed the PhD programme; responsible for the curriculum “Participatory development of AI models, methods and applications”
- 2023–25 **Member of the doctoral board**, *PhD in Economics*, University of Bergamo  
March 2023–April 2025

- 2023– **Member of the University of Bergamo’s Interdisciplinary Working Group on Artificial Intelligence (Tavolo di lavoro interdisciplinare sull’intelligenza artificiale dell’Università di Bergamo)**, University of Bergamo, (<https://www.unibg.it/node/16191>)
- 2023– **BSc Dissertation Officer**, University of Bergamo  
 Worked on improving procedures for supervisor allocation and on the technical integration of grade entry with ESSE3 (Cineca)
- Member of the working group on university guidelines for the adoption of artificial intelligence**, *Guidelines for students, faculty/researchers, and technical-administrative staff*, University of Bergamo
- Member of technical evaluation group**, *Procurement of university chatbot/LLM licences*, University of Bergamo
- Member of technical evaluation group**, *Procurement of a new student-services ticketing system*, University of Bergamo
- 2022–25 **TOLC examination commissioner**, *Three examination sessions*, University of Bergamo
- 2018–2022 **MSc Admission Officer**, *MSc programs: Operational Research, Operational Research & Finance, Operational Research & Statistics, Data and Decision Analytics*, University of Southampton
- 2017–18 **MSc Exam Officer**, *MSc programs: Operational Research, Operational Research & Finance, Operational Research & Statistics*, University of Southampton
- 2016–17 **Deputy Ethics Officer**, *Department of Mathematical Sciences*, University of Southampton

## Teaching innovation projects

**Project lead and budget holder**, *Tecnologie avanzate per la didattica nei corsi di informatica (86038) e di supporto ai corsi offerti dal Dipartimento di Scienze Economiche alla L-33*, University of Bergamo  
 Awarded through a competitive internal call

## Honors and awards

- 2021 **Paper in the “Selection of most cited papers” on EJCO**, paper title: “Bilevel programming methods for computing single-leader-multi-follower equilibria in normal-form and polymatrix games”, published on EURO Journal on Computational Optimization
- 2017 **STEM for Britain**, *Selected for “STEM for Britain” poster competition among early-career researchers at the House of Commons (Westminster), London, UK*
- 2008–10 **PhD scholarship**, *(1st classified over more than 80 applicants at Dipartimento di Elettronica e Informazione, Politecnico di Milano)*, granted by MIUR (Italian Ministry for University and Research)
- 2002–07 **BSc and MSc student scholarships**, *reduced student tuition fees (each eligible year) for merit*

## Teaching experience (as instructor/lecturer)

- 2025–26 **Neural Networks and Deep Learning**, *Instructor & module leader*, 48 hours, with  $\simeq 30$  MSc of Economics and Data Analysis, University of Bergamo  
 Topics: history of AI, foundations of neural networks and deep learning, backpropagation, feedforward neural networks, recurrent neural networks, convolutional and graph-covolutional neural networks, variational autoencoders, large language models and generative AI
- **Intelligenza Artificiale per la Comunicazione**, *Instructor & module leader*, 38 hours with  $\simeq 80$  Msc students of Comunicazione, Informazione ed Editoria, University of Bergamo  
 Topics: foundations of AI, history of AI, agentic modeling and reasoning, search problems and algorithms, neural networks and deep learning, large language models and generative AI
  - **INFORMATICA 86038**, *Instructor & module leader*, 48 hours with  $\simeq 350$  BSc students of Economics and Management, University of Bergamo  
 Topics: history of computer science, Boolean algebra and circuit synthesis, information representation, spreadsheets, databases, Python programming
  - **Soft & Digital Skills**, *Instructor*, 6 hours, BSc in Economics, University of Bergamo
  - **Soft & Computational Skills**, *Instructor*, 6 hours, MSc in Economics and Data Analysis, University of Bergamo
  - **Master Data Analyst per Decisioni Strategiche (DADS)**, *Instructor*, 8 hours, University of Bergamo  
 Topics: Intelligenza Artificiale: evoluzione, principi, strumenti
  - **Fondamenti Scientifico informatici dell’intelligenza artificiale (Scientific and Computer Science Foundations of Artificial Intelligence)**, *Instructor*, 20 hours, PhD in Artificial Intelligence for Sustainable Futures, University of Bergamo  
 Topics: overview of the key aspects of modern artificial intelligence and deep learning

- 2024–25 **Neural Networks and Deep Learning**, *Instructor & module leader*, 48 hours, with  $\simeq 30$  MSc of Economics and Data Analysis, University of Bergamo  
Topics: history of AI, foundations of neural networks and deep learning, backpropagation, feedforward neural networks, recurrent neural networks, convolutional and graph-covolutional neural networks, variational autoencoders, large language models and generative AI
- **Intelligenza Artificiale per la Comunicazione**, *Instructor & module leader*, 38 hours with  $\simeq 80$  Msc students of Comunicazione, Informazione ed Editoria, University of Bergamo  
Topics: foundations of AI, history of AI, agentic modeling and reasoning, search problems and algorithms, neural networks and deep learning, large language models and generative AI
  - **INFORMATICA 86038**, *Instructor & module leader*, 48 hours with  $\simeq 350$  BSc students of Economics and Management, University of Bergamo  
Topics: history of computer science, Boolean algebra and circuit synthesis, information representation, spreadsheets, databases, Python programming
  - **Soft & Digital Skills**, *Instructor*, 6 hours, BSc in Economics, University of Bergamo
  - **Soft & Computational Skills**, *Instructor*, 6 hours, MSc in Economics and Data Analysis, University of Bergamo
  - **Master Data Analyst per Decisioni Strategiche (DADS)**, *Instructor*, 4 hours, University of Bergamo  
Topics: artificial intelligence and generative AI
- 2023–24 **INFORMATICA 86038 & 87111**, *Instructor & module leader*, 48 hours (twice) with  $\simeq 150$  BSc students of Economics and Management, University of Bergamo  
Topics: history of computer science, Boolean algebra and circuit synthesis, information representation, spreadsheets, databases, Python programming
- 2022–23 **INFORMATICA 86038 & 87111**, *Instructor & module leader*, 48 hours (twice) with  $\simeq 150$  BSc students of Economics and Management, University of Bergamo  
Topics: history of computer science, Boolean algebra and circuit synthesis, information representation, spreadsheets, databases
- 2021–22 **MATH2014: Algorithms**, *Instructor & module leader*, 48 hours with  $\sim 80$  BSc students of Mathematics, University of Southampton  
Topics: Spanning Trees, Shortest Paths, Network Flows, Bipartite Matchings, Computational Complexity,  $\mathcal{NP}$ -Hardness and Approximability, Dynamic Programming  
., (sabbatical in Sem I)  
**Combinatorial Optimization, NATCOR**, *Course co-leader and co-instructor*,  $\sim 50$  PhD students, University of Southampton (virtual)  
Topics: Integer programming and norm minimization for machine learning, modeling techniques for integer programming, Lagrangian relaxations
- 2020–21 **MATH1058: Operational Research I and Mathematical Computing**, *Co-instructor & module leader*, 30 hours with  $\sim 160$  BSc students of Mathematics, University of Southampton  
Topics: Sorting Algorithms, Computational Complexity, Dijkstra's algorithm, Critical Path Method, Linear Programming, Simplex Method  
**MATH2014: Algorithms**, *Instructor & module leader*, 48 hours with  $\sim 80$  BSc students of Mathematics, University of Southampton  
Topics: Spanning Trees, Shortest Paths, Network Flows, Bipartite Matchings, Computational Complexity,  $\mathcal{NP}$ -Hardness and Approximability, Dynamic Programming
- 2019–20 **MATH1058: Operational Research I and Mathematical Computing**, *Co-instructor & module leader*, 30 hours with  $\sim 160$  BSc students of Mathematics, University of Southampton  
Topics: Sorting Algorithms, Computational Complexity, Dijkstra's algorithm, Critical Path Method, Linear Programming, Simplex Method  
**MATH2014: Algorithms**, *Instructor & module leader*, 48 hours with  $\sim 80$  BSc students of Mathematics, University of Southampton  
Topics: Spanning Trees, Shortest Paths, Network Flows, Bipartite Matchings, Computational Complexity,  $\mathcal{NP}$ -Hardness and Approximability, Dynamic Programming  
**Combinatorial Optimization, NATCOR**, *Co-instructor*,  $\sim 50$  PhD students, University of Southampton  
Topics: Integer programming and norm minimization for machine learning
- 2018–19 **MATH1058: Operational Research I and Mathematical Computing**, *Co-instructor & module leader*, 30 hours with  $\sim 160$  BSc students of Mathematics, University of Southampton  
Topics: Sorting Algorithms, Computational Complexity, Dijkstra's algorithm, Critical Path Method, Linear Programming, Simplex Method  
**MATH2014: Algorithms**, *Instructor & module leader*, 48 hours with  $\sim 80$  BSc students of Mathematics, University of Southampton  
Topics: Spanning Trees, Shortest Paths, Network Flows, Bipartite Matchings, Computational Complexity,  $\mathcal{NP}$ -Hardness and Approximability, Dynamic Programming

- 2017–18 **MATH1058: Operational Research I and Mathematical Computing**, *Co-instructor & module leader*, 30 hours with  $\sim 160$  BSc students of Mathematics, University of Southampton  
Topics: Sorting Algorithms, Computational Complexity, Dijkstra's algorithm, Critical Path Method, Linear Programming, Simplex Method
- MATH2014: Algorithms**, *Instructor & module leader*, 48 hours with  $\sim 80$  BSc students of Mathematics, University of Southampton  
Topics: Sorting Algorithms, Spanning Trees, Shortest Paths, Network Flows, Bipartite Matchings, Computational Complexity,  $\mathcal{NP}$ -Hardness and Approximability
- MATH2013: Introduction to Operational Research**, *Co-instructor & module co-leader*,  $\sim 60$  BSc students of Mathematics, University of Southampton  
Topics: Linear and Integer Programming, Simplex Method, Project Networks, Inventory Control
- 2016–17 **MATH2014: Algorithms**, *Instructor & module leader*, 48 hours with  $\sim 80$  BSc students of Mathematics, University of Southampton  
Topics: Sorting Algorithms, Spanning Trees, Shortest Paths, Network Flows, Bipartite Matchings, Computational Complexity,  $\mathcal{NP}$ -Hardness and Approximability
- 2015–16 **MATH2014: Algorithms**, *Co-instructor (with Prof. C. Potts)*, 24 hours with  $\sim 80$  BSc students of Mathematics, University of Southampton  
Topics: Network Flows, Bipartite Matchings, Computational Complexity,  $\mathcal{NP}$ -Hardness and Approximability
- 2015–16 **Optimization B**, *Co-instructor (with Prof. A. Koster)*, 20 hours with  $\sim 40$  BSc/MSc students of Mathematics, RWTH Aachen University  
Topics: Shortest paths, Linear Programming with TUM and TDI formulations, Bipartite and Nonbipartite Matchings
- 2011–12 **Foundations of Operations Research & module leader**, *Instructor*, 45 hours with  $\sim 100$  MSc students of Computer Science Engineering, Politecnico di Milano, Como Campus  
Topics: Shortest paths, Spanning Trees, Network flows, Linear Programming, Mixed-Integer Linear Programming

## Teaching experience (as teaching assistant)

- 2014–15 **Integer Linear Optimization**, *Teaching assistant (course tutor, homework grading)*, MSc students of Mathematics, RWTH Aachen University
- 2012–13 **Optimization B**, *Teaching assistant (course tutor, homework grading)*, BSc/master students of Mathematics, RWTH Aachen University
- 2010–11 **Optimization**, *Teaching assistant (course and lab demonstrations tutor)*, MSc students of Computer Science Engineering and Mathematical Engineering, Politecnico di Milano  
**Foundations of Operations Research**, *Teaching assistant (course and lab demonstrations tutor)*, MSc students of Computer Science Engineering and Mathematical Engineering, Politecnico di Milano
- 2009–10 **Optimization**, *Teaching assistant (course and lab demonstrations tutor)*, MSc students of Computer Science Engineering and Mathematical Engineering, Politecnico di Milano  
**Operations Research**, *Teaching assistant (course tutor)*, BSc students of Computer Science Engineering, Politecnico di Milano, Como campus
- 2008–09 **Fundamentals of Operations Research D**, *Teaching assistant (lab demonstrations tutor)*, BSc students of Computer Science Engineering, Politecnico di Milano, Como Campus  
**Fundamentals of Operations Research D-E**, *Teaching assistant (course tutor)*, BSc students of Computer Science Engineering, Politecnico di Milano
- 2007–08 **Fundamentals of Operations Research D**, *Teaching assistant (lab demonstrations tutor)*, BSc students of Computer Science Engineering, Politecnico di Milano, Como Campus  
**Complements of Operations Research**, *Teaching assistant (lab demonstrations tutor)*, MSc students of Computer Science Engineering and Mathematical Engineering, Politecnico di Milano  
**Fundamentals of Operations Research D-E**, *Teaching assistant (lab demonstrations tutor)*, MSc students of Computer Science Engineering, Politecnico di Milano

## Supervised students awards

- 2022 **CORMSIS Summer Project Prize**, *University of Southampton*, R. Anand
- 2021 **CORMSIS Summer Project Prize**, *University of Southampton*, Q. Du
- 2020 **CORMSIS Summer Project Prize**, *University of Southampton*, M.E. Voss, A. Inaba, X.-W. Teh
- 2019 **CORMSIS Summer Project Prize**, *University of Southampton*, S.Y. Chin and Z. Xue
- 2018 **CORMSIS Summer Project Prize**, *University of Southampton*, M. Orderud
- 2017 **CORMSIS Summer Project Prize**, *University of Southampton*, C. Velez

## Supervision of postdoctoral researchers

- 2024–26 **G. Andreana**, *University of Bergamo*  
2024–25 **C. Tomasi**, *University of Bergamo*  
2020–22 **F. Garuba**, *University of Southampton*  
2020–20 **P.M. Bartmeyer**, *University of Southampton*

## Supervision of PhD students (graduated)

- 2022–2026 **D. Benfield**, *University of Southampton*  
(co-supervision (33%) with A. Zemkoho and V. Phan)  
2022–2026 **J. Montree**, *University of Southampton*  
(co-supervision (50%) with H. Qi)  
2019–2023 **M.A. Alammar**, *University of Southampton*  
(co-supervision (30%) with C. Potts and A. Martinez-Sykora)  
2018–2023 **A.J. Dunn**, *University of Southampton*  
(co-supervision (40%) with A. Zemkoho and H. Qi)  
2018–2022 **J.B. Badrodin**, *University of Southampton*  
(co-supervision (50%) with H. Qi)  
2017–2022 **K.-M. Steinborn-Büsse**, *University of Southampton*  
(co-supervision (50%) with J. Fliege)  
2016–21 **R. Walton**, *University of Southampton*  
(co-supervision (50%) with J. Fliege)

## Supervision of PhD students (ongoing)

- 2025– **F. Belotti**, *University of Bergamo*  
PhD in Artificial Intelligence for Sustainable Futures

## MSc/BSc student supervision (dissertations)

- 2022– **4 BSc students**, *University of Bergamo*  
(single supervisor)  
2021–22 **5 MSc students**, *University of Southampton*, R. Anand, A. Arora, L. Main, S. Crofts, J. Rapley  
(single supervisor)  
2020–21 **7 MSc students**, *University of Southampton*, Z. Chen, Q. Du, R. Jain, L. Moore, T. Boivert, S. Wang, T. Wu  
(single supervisor)  
2019–20 **4 MSc students**, *University of Southampton*, M. Maliali, A. Inaba, X.-W. Teh, M.E. Voss  
(single supervisor)  
2018–19 **1 Visiting MSc student**, *University of Southampton/University of Pavia*, L. Muffone  
(single supervisor)  
**3 MSc students**, *University of Southampton*, S.-Y. Chin, Z. Fei, Z. Xue  
(single supervisor)  
2017–18 **3 MSc students**, *University of Southampton*, M. Orderud, Z. Luu, Q. Wu  
(single supervisor)  
2016–17 **3 MSc students**, *University of Southampton*, C. Liu, G. Oyasor, C. Velez  
(single supervisor)  
2015–16 **2 MSc students**, *University of Southampton*, K. Roy, C. Ma  
(single supervisor)  
2009–16 **3 MSc students**, *Politecnico di Milano*, A. Nicotra, C. Ileri, D. Cardana  
(co-supervision with E. Amaldi)  
2015 **1 MSc student**, *RWTH Aachen University*, J. Rosendahl  
(co-supervision with A. Koster)  
2009–16 **3 MSc students**, *Politecnico di Milano*, A. Marchesi, S. Corti, A. Perrucci  
(co-supervision with N. Gatti)



2008–12 **7 BSc students**, *Politecnico di Milano*, A. Nicotra, M. Mazzoni, G. Personeni, I. Catallo, S. Celentano, F. Cardellini, E. Cereda  
(co-supervision with E. Amaldi)

## Students supervision (within taught courses)

2020–21 **MATH6145: Presenting Reports**, University of Southampton  
Supervisor of 10 MSc students on researching, writing, and presenting a report on aspects of Operations Research and Machine Learning

2020–21 **MATH3092: Mathematics Project (MMATH)**, University of Southampton  
Supervisor of a 4th year BSc student (A. Rose) working on solving the hyperplane-clustering problem

2020–21 **MATH3092: Mathematics Project**, University of Southampton  
Supervisor of six BSc students working on modeling and solving diet problems with mathematical programming

2019–20 **MATH6145: Presenting Reports**, University of Southampton  
Supervisor of six MSc students on researching, writing, and presenting a report on aspects of Operations Research and Machine Learning

2018–19 **MATH3032: Mathematical Investigation and Communication (MIC)**, University of Southampton  
Supervisor of six BSc students working on modeling and solving diet problems with mathematical programming

2017–18 **MATH6145: Presenting Reports**, University of Southampton  
Supervisor of three MSc students on researching, writing, and presenting a report on aspects of Operations Research

**MATH3032: Communicating and Researching Mathematics (CRM)**, University of Southampton  
Supervisor of four BSc students working on topics of network traffic estimation

2016–17 **MATH6145: Presenting Reports**, University of Southampton  
Supervisor of three MSc students on researching, writing, and presenting a report on aspects of Operations Research

**MATH3032: Communicating and Researching Mathematics (CRM)**, University of Southampton  
Supervisor of three BSc students working on topics of Robust Optimization

2015–16 **MATH6145: Presenting Reports**, University of Southampton  
Supervisor of three MSc students on researching, writing, and presenting a report on aspects of Operations Research

## MPhil/PhD students examination

2026 **PhD examination committee member**, *University of Parma*, Information Technologies, 38th cycle; final examination committee

2024 **PhD examination committee member**, *University of Pavia*, Mathematics; final examination of S. Milanese and M.L. Trezzi

2024 **PhD examination committee member**, *University of Pavia*, Computational Mathematics, Learning and Data Science; final examination committee

2024–25 **PhD examination committee member**, *TU Berlin*, L. Eifler (PhD student in Mathematics)

2022–23 **PhD examination committee member**, *University of Milano-Bicocca*, D. Marelli, F. Adobbati, L. Guglielmo, E. Kurian, M. Saletta (PhD students in Computer Science)

2022–23 **External examiner for PhD thesis viva/defense**, *University of Pavia*, A. Codegoni (PhD student in Computational Mathematics and Decisions Sciences)

2021–22 **External examiner for PhD thesis viva/defense**, *University of Technology of Compiegne*, C. Wang (PhD student in Informatics)

**Internal examiner for MPhil thesis viva/defense**, *University of Southampton*, H. Hall (MPhil student in Operational Research)

**Independent chair for PhD thesis viva/defense**, *University of Southampton*, G. Yim (PhD student in Applied Mathematics and Theoretical Physics)

2020–21 **Internal examiner for PhD thesis viva/defense**, *University of Southampton*, Y. Zhang (PhD student in Operational Research)

2019–20 **Internal examiner for PhD thesis viva/defense**, *University of Southampton*, M. Benedek (PhD student in Operational Research)

2018–19 **Internal examiner for PhD thesis viva/defense**, *University of Southampton*, A. Robins (PhD student in Operational Research)

**Internal examiner of PhD thesis viva/defense**, *University of Southampton*, S. Zhou (PhD student in Operational Research)

**Independent assessor for 1st year review**, *University of Southampton*, L.-C. Dinh (PhD student in Computer Science)

**Independent assessor for 3rd year review**, *University of Southampton*, F. Ge (PhD student in Operational Research)

2017–18 **Internal examiner for PhD thesis viva/defense**, *University of Southampton*, C. Lamas Fernandez (PhD student in Business Analytics)

**Independent assessor for 2nd year review (MPhil to PhD upgrade)**, *University of Southampton*, Y. Zhang (PhD student in Operational Research)

**Independent assessor for 2nd year review (MPhil to PhD upgrade)**, *University of Southampton*, S. Zachariades (PhD student in Operational Research)

**Independent assessor for 2nd year review (MPhil to PhD upgrade)**, *University of Southampton*, F. Ge (PhD student in Operational Research)

2016–17 **Independent assessor for 2nd year review (MPhil to PhD upgrade)**, *University of Southampton*, M. Benedek (PhD student in Operational Research)

**Independent assessor for 2nd year review (MPhil to PhD upgrade)**, *University of Southampton*, S. Zhou (PhD student in Operational Research)

**Independent assessor for 1st year PhD review**, *University of Southampton*, Y. Zhang (PhD student in Operational Research)

**Independent assessor for annual PhD review**, *University of Southampton*, H. Hall (PhD student in Operational Research)

2015–16 **Independent assessor 2nd year review (MPhil to PhD upgrade)**, *University of Southampton*, A. Robins (PhD student in Operational Research)

2011–12 **Member of MSc graduation committee**, *Politecnico di Milano*, Mathematical Engineering

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## Spoken languages

Italian (mother tongue), English (fluent), French (basic)

September 9, 2026  
Dr. Stefano Coniglio

